MAT2540, Classwork16, Spring2026

10.3 Representing Graphs and Graph Isomorphism Graphs (Conti.)

7. Observations of the Adjacency Matrix For a Directed Graph

Loops: If there is a loop, it shows on the diagonal of the matrix as "1"

Sum of the row/column: sums of rows represent out-degree, sums of columns represent in-degree

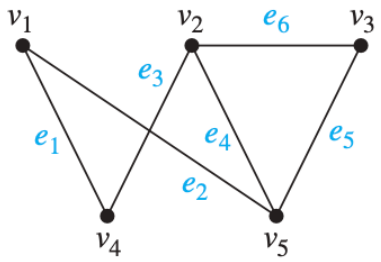
Symmetric property: NO, it is NOT symmetric since lower triangle of the matrix is not mirroring the upper triangle of the matrix.

8. *Definition:* **Incidence Matrix** based on the Incidence of Vertices and Edges of an **Undirected Graph**.

Let $G = (V, E)$ be an undirected graph. Suppose that $v_1, v_2, \ldots, v_n$ are the vertices and $e_1, e_2, \ldots, e_m$ are the edges of G. Then the incidence matrix with respect to this ordering of V and E is the $n \times m$ matrix is $M = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i, \\ 0 & \text{otherwise} \end{cases}.$$

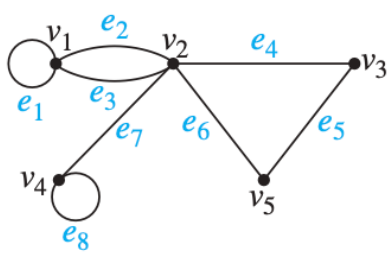
9. Example. Represent the given graph with an incidence matrix of an undirected simple graph.



$$\begin{array}{c|cccccc|c} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & \\ \hline v_1 & 1 & 1 & 0 & 0 & 0 & 0 & 2 \\ v_2 & 0 & 0 & 1 & 1 & 0 & 1 & 3 \\ v_3 & 0 & 0 & 0 & 0 & 1 & 1 & 2 \\ v_4 & 1 & 0 & 1 & 0 & 0 & 0 & 2 \\ v_5 & 0 & 1 & 0 & 1 & 1 & 0 & 3 \end{array}$$

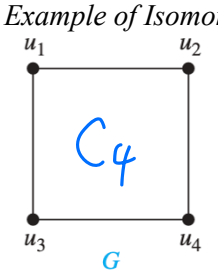
sum of row = deg (of vertex)

10. Example. Represent the given graph with an incidence matrix of a pseudograph.



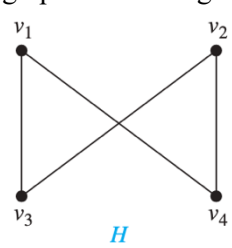
$$\begin{array}{c|cccccccc} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \hline v_1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ v_2 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ v_3 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ v_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ v_5 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}$$

11. *Example of Isomorphism*. Find the Adjacency Matrix of each given graph and the degrees of each vertex.



$$\begin{array}{c|cccc} & u_1 & u_2 & u_3 & u_4 \\ \hline u_1 & 0 & 1 & 1 & 0 \\ u_2 & 1 & 0 & 0 & 1 \\ u_3 & 1 & 0 & 0 & 1 \\ u_4 & 0 & 1 & 1 & 0 \end{array}$$

vertex	u_1	u_2	u_3	u_4
degree	2	2	2	2



bipartite

$$\begin{array}{c|cccc} & v_1 & v_2 & v_3 & v_4 \\ \hline v_1 & 0 & 0 & 1 & 1 \\ v_2 & 0 & 0 & 1 & 1 \\ v_3 & 1 & 1 & 0 & 0 \\ v_4 & 1 & 1 & 0 & 0 \end{array}$$

vertex	v_1	v_2	v_3	v_4
degree	2	2	2	2

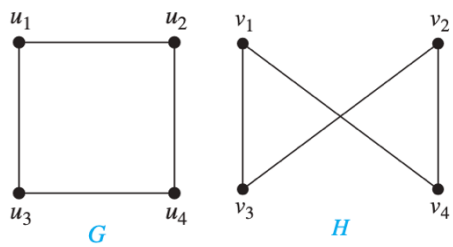
Are these two graphs "the same"?

12. *Definition:* **Isomorphism** of Graphs.

The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there exists a one-to-one and onto function f from V_1 to V_2 with the property that u and v are adjacent in G_1 if and only if $f(u)$ and $f(v)$ are adjacent in G_2. Such a function f is called isomorphism.

Two simple graphs that are not isomorphic are called non isomorphic.

13. Find the **isomorphism** f from definition in 12 to show that the graphs $G = (V, E)$, $H = (W, F)$ are isomorphic.



$$A_G = \begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \\ u_4 \end{array} \begin{array}{c} \begin{array}{cccc} u_1 & u_2 & u_3 & u_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array} \qquad A_H = \begin{array}{c} \\ v_1 \\ \boldsymbol{v_4} \\ v_3 \\ \boldsymbol{v_2} \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & \boldsymbol{v_4} & v_3 & \boldsymbol{v_2} \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

The function f with $f(u_1)=v_1$, $f(u_2)=v_4$, $f(u_3)=v_3$, $f(u_4)=v_2$ is a one-to-one and onto function. which is an isomorphism

14. *Definition:* **Graph Invariant**

A property preserved by isomorphism of graphs is called a graph invariant. The isomorphic simple graphs need

(1) Same number of vertices: Because there is a one-to-one correspondence between the sets of vertices of the graphs.

(2) Same number of edges Because the one-to-one correspondence between vertices establishes a one-to-one correspondence between edges.

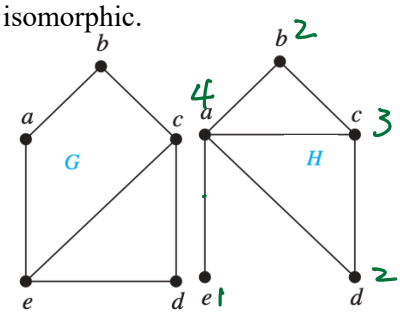
(3) The degree sequence of the vertices in isomorphic simple graphs must be the same.

If (1), (2), and (3) are satisfied, then check if the isomorphism f exists or not

It might be hard to prove two graphs are isomorphic, but it is easier to show two graphs are not isomorphic.

15. *Example.* Check if an **isomorphism** f exists for the given graphs $G = (V, E)$, $H = (W, F)$.

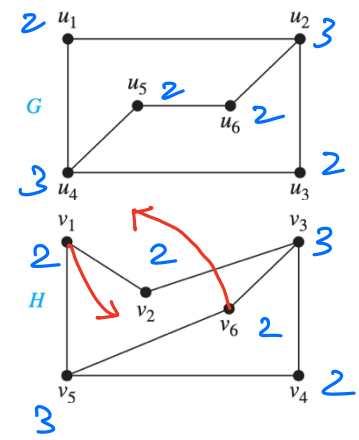
The Graph	The number of vertices	The number of edges	degree sequence
G	$\lvert V\rvert =$ 5	$\lvert E\rvert =$ 6	3, 3, 2, 2, 2
H	$\lvert W\rvert =$ 5	$\lvert F\rvert =$ 6	4, 3, 2, 2, 1



Conclusion: NO, G and H are not isomorphic.

16. *Example.* Check if an **isomorphism** f exists for the given graphs $G = (V, E)$, $H = (W, F)$.

The Graph	The number of vertices	The number of edges	degree sequence
G	$\lvert V\rvert =$ 6	$\lvert E\rvert =$ 7	3, 3, 2, 2, 2, 2
H	$\lvert W\rvert =$ 6	$\lvert F\rvert =$ 7	3, 3, 2, 2, 2, 2



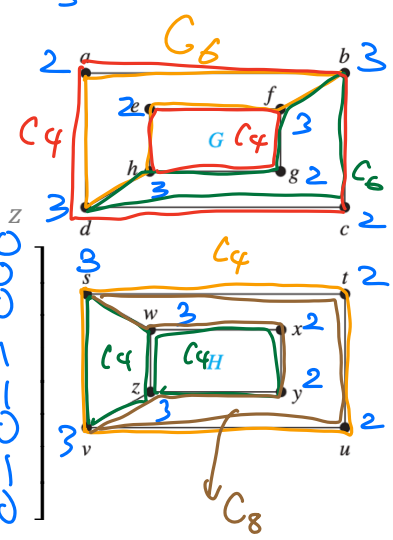
$$\begin{array}{c} \\ u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{array} \begin{array}{c} \begin{array}{cccccc} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array} \qquad \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} \begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$f(u_1) = v_6$
$f(u_2) = v_3$
$f(u_3) = v_4$
$f(u_4) = v_5$
$f(u_5) = v_1$
$f(u_6) = v_2$

Conclusion: Yes, it's isomorphism

17. *Example.* Check if an **isomorphism** f exists for the given graphs $G = (V, E)$, $H = (W, F)$.

The Graph	The number of vertices	The number of edges	degree sequence
G	$\lvert V\rvert =$ 8	$\lvert E\rvert =$ 10	3, 3, 3, 3, 2, 2, 2, 2
H	$\lvert W\rvert =$ 8	$\lvert F\rvert =$ 10	3, 3, 3, 3, 2, 2, 2, 2



$$\begin{array}{c} \\ a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{array} \begin{array}{c} \begin{array}{cccccccc} a & b & c & d & e & f & g & h \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array} \qquad \begin{array}{c} \\ s \\ t \\ u \\ v \\ w \\ x \\ y \\ z \end{array} \begin{array}{c} \begin{array}{cccccccc} s & t & u & v & w & x & y & z \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

Conclusion: NO, it is not isomorphic.